VECTOR

MICROSAR Ethernet Transceiver Driver

Technical Reference

MII Transceivers

Version 2.0.0

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Status	Released

Document Information

History

Author	Date	Version	Remarks
Feßler, David	2014-12-02	1.0.0	Creation of document
Mark Harsch	2015-03-11	1.1.0	▶ ESCAN00081256: FEAT-705: Ethernet wakeup based on Activation Line [AR4-1006] ▶ ESCAN00080870: Rename Technical Reference according to SPEC-63403
Feßler, David	2015-05-27	1.2.0	▶ ESCAN00083155: Add support for extended register access and TransceiverInit-Callouts
David Röder	2017-04-18	1.3.0	▶ ESCAN00094790: Removed all transceiver driver general aspects
Amr Abdelhady	2018-12-4	1.4.0	▶ STORYC-6225: Update subpackages to DrvTrans__coreEthAsr v. 3.00.xx

Table 1-1 History of the document

Reference Documents

No.	Title	Version
[1]	TechnicalReference_EthTrcv_Core	See Delivery

Table 1-2 Reference documents

Scope of the Document

This complement to the technical reference [1] describes the derivative dependent aspects of the Ethernet Transceiver Driver basis software. Please refer to your Release Notes to get a detailed description of the platform (host, compiler, and transceiver) your Vector Ethernet Bundle has been configured for.



Please note
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.



Note

This document describes aspects that apply for Ethernet Transceiver Driver modules provided by Vector, which are only used for standard MII transceivers. For this reason the corresponding infixes of APIs, macros, parameters etc. are replaced by Ethmii or, if only uppercase is used, ETHMII.

The derivative independent aspects of the Ethernet Transceiver Driver are discussed in [1], where the above mentioned infix is replaced by the markers <Derivative> and <DERIVATIVE>.

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Component History

Component Version	New Features
01.00.00	created
01.01.xx	Wakeup Support per Activationline according to AUTOSAR 4.2.1
01.02.xx	Extended register access and EthTrcv_TransceiverInit-callouts
02.00.xx	MICROSAR 4 Release 14
03.00.xx	MICROSAR 4 Release 17
04.00.xx	Ethernet Transceiver Core Introduced
04.01.xx	FEAT-2234: Ethernet Link Quality Monitoring
05.00.xx	STORYC-6225: Update subpackages to DrvTrans__coreEthAsr v. 3.00.xx

Table 1-3 Component history

1 Introduction

This document describes the derivative dependent functionality and configuration of the Ethernet Transceiver Driver.

Supported AUTOSAR Release:	4.1.1	
Supported Configuration Variants:	pre-compile	
Vendor ID:	ETHTRCV_30_ETHMII_VENDOR_ID	30 decimal (=Vector-Informatik, according to HIS)
Module ID:	ETHTRCV_30_ETHMII_MODULE_ID	73 decimal

The Ethernet Transceiver Driver provides hardware independent access to control connected transceivers in a generic way. It offers the functionality to control the mode of operation of connected transceivers as well as to determine their current state, e.g. if events like link status change or bus errors happened.

The transceiver itself is a hardware device, which mainly transforms the logical I/0 signals of the Ethernet Controller to the bus compliant electrical levels, currents and timings.

The transceiver driver considered in this technical reference is designed to operate with all standard MII transceivers. This documentation however only covers the special aspects of the driver software covering this kind of transceivers, while all general aspects are covered by the complementary document [1].

2 Hardware Overview

Table 2-1 summarizes the supported derivatives and settings this Ethernet Transceiver Driver can be used on. It also provides information on the documentation used during development of the driver for detailed reference to hardware specific information.

Derivatives	▶ IEEE 802.3 conformant 10BASE-T/100BASE-TX/1000BASE-T transceivers connected with MII.
Compiler	Compiler independent
Hardware Manuals	▶ DP83848Q-Q1 PHYTER Datasheet Automotive 10 BASE-T, 100 BASE-TX Transceiver Doc. No. SNLS341C Rev. N/A March 23, 2015 ▶ 88EA1517 Datasheet Automotive 10/100 BASE-T, 100 BASE-TX Transceiver Doc. No. MV-S110945-00 Rev. N/A ▶ 88EA1512 Datasheet Automotive 10/100/1000 BASE-T, 1000 BASE-X Transceiver Doc. No. MV-S110945-00 Rev. N/A February 2016 ▶ LAN8710A/LAN8710Ai Datasheet Automotive 10 BASE-T, 100 BASE-TX Transceiver Doc. SMSC LAN8710/LAN8710Ai Rev. 1.4 August 2012
Safety Manuals	-
Errata Sheets	-
Specifics	-
Used Registers	See EthTrcv_30_Ethmii_LL_Regs.h
Hardware features related to independence or partitioning	-
Access Mechanism	▶ SMI management interface by utilizing Eth or EthSwt ▶ Ethernet switch internal registers by utilizing EthSwt
Hardware Diagnostics	▶ Internal Loopback mode

Table 2-1 Hardware Overview

3 Derivative Specific Functional Description

3.1 Overview

The general functional description of the transceiver driver is part of the general technical reference [1]. This section gives an overview over which of the features are supported by the specific transceiver driver considered in this document and to what extent. The features marked with ■ in Table 3-1 are available for the Ethernet transceiver driver for standard MII transceivers. If a feature is supported with limitations, it is marked with ■. The limitations are explained in section 3.2. If a feature is not supported, it is marked with ■.

Supported AUTOSAR Standard Conform Features	
Initialization of the Ethernet Transceiver	■
Setting and getting the Transceiver Mode	■
Transceiver wakeup	■
Restart auto negotiation	■
Get link state	■
Get baud rate	■
Get duplex mode	■
Set test mode	■
Set loopback mode	■
Get signal quality	■
Set TX mode	■
Get cable diagnostics result	■
Get PHY identifier	■
Retrieve version info	■
Report development errors to DET	■
Report production errors to DEM	■

Table 3-1 Supported AUTOSAR Standard Conform Features

3.2 Limitations to supported Features and Functions

3.2.1 PHY Test Modes

The PHY test modes are explained in detail in the general technical reference [1]. In Table 3-2 all test modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

TestMode value	
ETHTRCV_PHYTESTMODE_NONE	■
ETHTRCV_PHYTESTMODE_1	■
ETHTRCV_PHYTESTMODE_2	■

ETHTRCV_PHYTESTMODE_3	■
ETHTRCV_PHYTESTMODE_4	■
ETHTRCV_PHYTESTMODE_5	■

Table 3-2 Supported PHY Test Modes

3.2.2 PHY Loopback Modes

The PHY loopback modes are explained in detail in the general technical reference [1]. In Table 3-3 all loopback modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

LoopbackMode value	
ETHTRCV_PHYLOOPBACK_NONE	■
ETHTRCV_PHYLOOPBACK_INTERNAL	■
ETHTRCV_PHYLOOPBACK_EXTERNAL	■
ETHTRCV_PHYLOOPBACK_REMOTE	■

Table 3-3 Supported PHY Loopback Modes

3.2.3 PHY TX Modes

The TX modes are explained in detail in the general technical reference [1]. In Table 3-4 all TX modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

TxMode value	
ETHTRCV_PHYTXMODE_NORMAL	■
ETHTRCV_PHYTXMODE_TX_OFF	■
ETHTRCV_PHYTXMODE_SCRAMBLER_OFF	■

Table 3-4 Supported PHY Tx Modes

3.3 Extended Register Access

This driver is designed to support all transceivers with standard MII register set. Setting hardware dependent options like enabling RMII mode is not possible with a generic driver because access to vendor specific registers is needed and these registers are unknown for a generic driver.

Extended Register Access enables the user to configure a hardware dependent register address and a corresponding register value which is written to this address at initialization time. Using this mechanism a setting like RMII mode or extended initialization can be done to support transceivers which need access to non-standard MII-registers.

To enable additional register access during transceiver initialization, a register address and a value to write (16 bit) into that register can be specified, as can be seen in Figure 3-1.

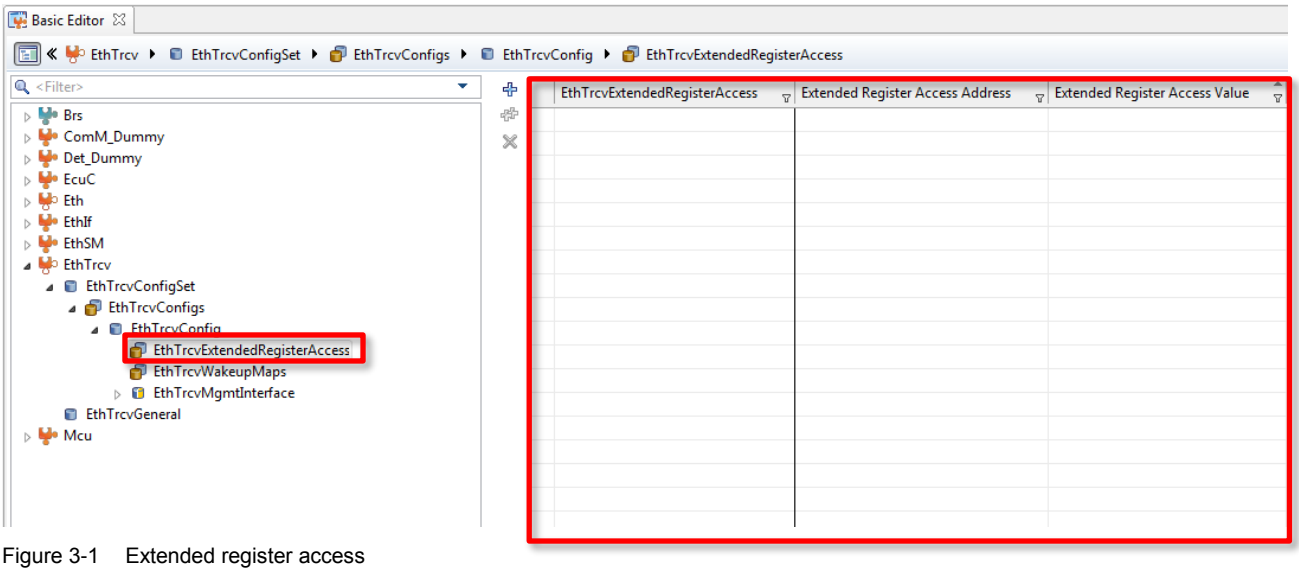


Figure 3-1 Extended register access

Note

Add a `EthTrcvExtendedRegisterAccess` containers to write a vendor specific register at initialization time

3.4 Transceiver Access Check

For integration and test purposes an MDIO access check can be configured at startup. In the method `EthTrcv_30_Ethmii_TransceiverInit()` the constant MII register 0x02 (PHY identifier 1) is read and checked for a preconfigured value, which can be obtained from the transceiver‘s datasheet. The configuration parameters are shown in Figure 3-2.

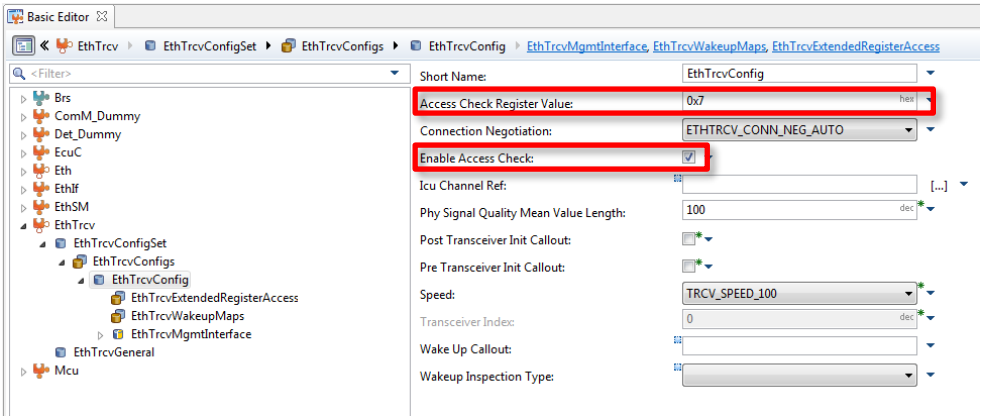


Figure 3-2 Transceiver Configuration: Access check

4 Glossary and Abbreviations

4.1 Abbreviations

Abbreviation	Description
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
BSW	Basis Software
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DIO	Digital Input Output
EAD	Embedded Architecture Designer
ECU	Electronic Control Unit
EcuM	ECU Manager
Eth	Ethernet Controller Driver
EthIf	Ethernet Interface
EthTrcv	Ethernet Transceiver Driver
HIS	Hersteller Initiative Software
MDIO	Management Data Input Output
MICROSAR	Microcontroller Open System Architecture (the Vector AUTOSAR solution)
Platform	Hardware including host and communication controller (might also be integrated in host) on which the communication stack is implemented.
RTE	Runtime Environment
SRS	Software Requirement Specification
SWC	Software Component
SWS	Software Specification

Table 4-1 Abbreviations

5 Contact

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